Verification appendix

2 November 2026

Companion to: Quotient-Group Obstructions for Pure Type II Divisor Constructions in the Erdős–Straus Equation.

1. Reproduction and logical scope

The accompanying archive contains `verify.py`, the short certificate printed in the manuscript, and three deterministic JSON outputs. Python 3.8 or later is sufficient; only the standard library is used. Extract the archive and run:

```
python3 verify.py -output-dir rerun
python3 printed_certificate.py
```

The first command writes three files: `modulus_certificates.json`, `family_certificates.json`, and `summary.json`. They should agree with the supplied copies in `certificates/`. The second command exits successfully without output. Run it without Python's `-O` option, which disables its assertions. The larger verifier uses an explicit check function that remains active with optimization.

All arithmetic is integer arithmetic, except for exact `Fraction` objects used to check symbolic coefficients. Primality is decided by trial division through the integer square root. No probabilistic tests, floating-point estimates, external mathematics packages, or network access are required.

What is proved outside the code. The manuscript proves the quotient criterion, the necessary index at least eight, and its base-2 equality classification. These are prerequisites for the finite exclusions; the code does not infer the universal theorems from samples. The lower bounds at 11 and 31 also have analytic proofs. The code redundantly checks their complete smaller-modulus subgroup domains.

Claim	Complete domain checked
Minimum 11	Every unit-generated cyclic subgroup at $a = 3, 7$.
Minimum 31, $H \neq \{1\}$	Every nontrivial unit-generated cyclic subgroup at $a = 3, 7, 11, 15, 19, 23, 27$.
Base-2 minimum 127	All 31 moduli $a = 4c - 1$, $1 \leq c \leq 31$.
Composite base-2 minimum 195	All 48 eligible $a = 4c - 1$, $1 \leq c \leq 48$, classified by exact primality; retain all 25 composites.

For the latter two domains every computed quotient index is at most six. This excludes *all* choices of the other prime bases and their positive exponents by the manuscript's index theorem. There is no hidden prime cap or exponent cap. Explicit witnesses attain each modulus. No minimum prime product, uniqueness classification, or four-dimensional result is certified here.

The program independently compares the totient obtained by unit enumeration with the factorization formula, builds actual power-generated subgroups, partitions the full unit group into actual cosets, and tests the complete finite ordinary and doubled residual exponent boxes. Every numerical identity in the manuscript is checked against the original fraction equation.

2. Complete lower-modulus arithmetic

Here $H = \langle 2 \rangle$ modulo a, $L = |H| = \mathrm{ord}_a(2)$, and $Q = (\mathbb{Z}/a\mathbb{Z})^\times / H$. Actual subgroup elements and the primality classification of every candidate are recorded in the JSON. The two tables intentionally overlap.

All eligible $a < 127$ (31 rows)

a	$\varphi(a)$	L	$\lvert Q\rvert$	$-1 \in H$
3	2	2	1	yes
7	6	3	2	no
11	10	10	1	yes
15	8	4	2	no
19	18	18	1	yes
23	22	11	2	no
27	18	18	1	yes
31	30	5	6	no
35	24	12	2	no
39	24	12	2	no
43	42	14	3	yes
47	46	23	2	no
51	32	8	4	no
55	40	20	2	no
59	58	58	1	yes
63	36	6	6	no
67	66	66	1	yes
71	70	35	2	no
75	40	20	2	no
79	78	39	2	no
83	82	82	1	yes
87	56	28	2	no
91	72	12	6	no
95	72	36	2	no
99	60	30	2	yes
103	102	51	2	no
107	106	106	1	yes
111	72	36	2	no
115	88	44	2	no
119	96	24	4	no
123	80	20	4	no

Composite $a < 195$ (25 rows)

a	$\varphi(a)$	L	$\lvert Q\rvert$	$-1 \in H$
15	8	4	2	no
27	18	18	1	yes
35	24	12	2	no
39	24	12	2	no
51	32	8	4	no
55	40	20	2	no
63	36	6	6	no
75	40	20	2	no
87	56	28	2	no
91	72	12	6	no
95	72	36	2	no
99	60	30	2	yes
111	72	36	2	no
115	88	44	2	no
119	96	24	4	no
123	80	20	4	no
135	72	36	2	no
143	120	60	2	no
147	84	42	2	no
155	120	20	6	no
159	104	52	2	no
171	108	18	6	yes
175	120	60	2	no
183	120	60	2	no
187	160	40	4	no

Interpretation. Every row has $|Q| < 8$, regardless of whether the sign coset is nontrivial. No prime-support or exponent choices remain to be searched at these moduli. The index inequality alone excludes an eventual pure family.

3. Complete quotient witnesses at the four bounds

For each row below, $x_m = q_1^m q_2^r q_3^s$, $A = q_2 H$, $B = q_3 H$, $N = -H$, and $T = (-4^{-1})H$. The finite conditions checked are

$$N \notin \{A^i B^j : 0 \le i \le r, 0 \le j \le s\}, \quad T \notin \{A^i B^j : 0 \le i \le 2r, 0 \le j \le 2s\},$$

together with some $A^u = NB^v$, $1 \le u \le r$, $1 \le v \le s$. The listed integer shift lifts this relation as $q_1^\delta q_2^u \equiv -q_3^v \pmod{a}$.

a	q_1	q_2	q_3	(r,s)	$\lvert H\rvert$	$\lvert Q\rvert$	(u,v)	δ	m_0
11	23	7	37	$(1,1)$	1	10	$(1,1)$	0	1
31	5	7	11	$(1,1)$	3	10	$(1,1)$	-1	1
127	2	3	11	$(1,2)$	7	18	$(1,2)$	1	1
195	2	11	19	$(1,1)$	12	8	$(1,1)$	4	4

The lifts follow from $7 + 37 = 44$, $7 + 5 \cdot 11 = 62$, $2 \cdot 3 + 11^2 = 127$, and $16 \cdot 11 + 19 = 195$. In the second equality multiply the congruence by 5^{-1} to obtain $\delta = -1$.

Actual subgroups.

$$\begin{aligned} H_{11} &= \{1\}, \\ H_{31} &= \{1, 5, 25\}, \\ H_{127} &= \{1, 2, 4, 8, 16, 32, 64\}, \\ H_{195} &= \{1, 2, 4, 8, 16, 32, 49, 61, 64, 98, 122, 128\}. \end{aligned}$$

The program records all cosets and their orders. The $a = 11$ and $a = 31$ doubled residual residue sets printed in the manuscript have eight and nine elements, respectively; at 31 the latter represent eight cosets because $18H = 28H$. No distinct-coset count is inferred from the number of residue entries.

Modulus 127. The coset $3H$ generates a cyclic quotient of order 18. In additive labels, $11H = 14$ and $N = T = 9$. The doubled box has 15 exponent pairs but 14 distinct cosets, omitting exactly $\{5, 9, 13, 17\}$. The program verifies each label using actual residue cosets.

Modulus 195 and sharpness. With logarithms to base 2 in the unit groups modulo $3, 5, 13$, define

$$\Psi(g) = (\alpha(g) - \gamma(g) \bmod 2,\ \beta(g) - \gamma(g) \bmod 4).$$

The program checks all 96 units and all $96^2 = 9216$ multiplication pairs: Ψ is a surjective homomorphism to $C_2 \times C_4$ with kernel H_{195}. The images are $\Psi(-1) = (1, 0)$, $\Psi(11) = (0, 1)$, $\Psi(19) = (1, 1)$. Both fixed cosets have order four, N is not a square, and the doubled box is exactly $Q \setminus \{N\}$. This is a finite structural certificate for the existing sharpness example, not a numerical sample of an infinite assertion.

4. Exact examples, samples, and the source correction

Every point check enumerates all divisors from the known prime factorization and tests both forbidden ordinary targets, the forbidden quadratic Type I target, and the Type II witnesses. For each included completion it verifies $ed = x^2$, $e, d \nmid x$, integral positive denominators, and

$$4xyz = p(xy + xz + yz).$$

The following samples use $y = p(x + d)/a$ and $z = p(x + e)/a$, consistently with the manuscript; interchanging y, z changes no mathematical statement.

a	m	p	x	y	z
31	2	7669	1925	4217950	536830
127	1	2777	726	336017	16662
127	3	11489	2904	5560676	275736
195	5	26557	6688	1009166	9348064
195	6	53309	13376	4051484	37529536
195	8	213821	53504	65001584	602119936

All p values in this table are prime by deterministic trial division. The first-point examples for moduli 11 and 31 have composite $p = 23817$ and 1509, respectively, and are used only as modular witnesses.

Precisely bounded sampling. Complete prime-construction exponent lists within the tested intervals are:

a	Point range checked	Prime, pure exponents in that range
11	$1 \le m \le 4$	none
31	$1 \le m \le 6$	$\{2\}$
127	$1 \le m \le 6$	$\{1, 3, 5, 6\}$
195	$1 \le m \le 8$	$\{5, 6, 8\}$

At 195, $m = 1, 2, 3$ have no Type II square-divisor hit; the family starts at $m = 4$. The code checks these three prethreshold values separately. The universal modular conclusions come from the manuscript's quotient proof and lifted formulas, not from the sample ranges. No assertion about infinitely many prime values is made.

Remaining ledger typo found during this preparation. The active reconciliation's numerical correction paragraph (extraction label A12G0788, Section 3.7 bullet for $m = 8$) gives $y = 64{,}998{,}584$ for the modulus-195 family. Its own other parameters imply

$$y = \frac{213821(53504 + 5776)}{195} = 19 \cdot 2^4 \cdot 213821 = 65{,}001{,}584.$$

For the incorrect y, the integer residual $4xyz - p(xy + xz + yz)$ equals $-317{,}919{,}326{,}208{,}000$; for the corrected value it is zero. The verifier records both. This error changes neither the symbolic family nor the minimum-modulus proof. The master ledger remains unmodified.

Source and reproducibility record. The mathematical input was the live ledger revision reconciled on 2 November 2026 (Drive modification time 15:40:31 UTC), read alongside the 1 November referee report and Assignment 12 prior-art report/search record. A manifest in the archive records the precise document ID and revision, file hashes, and verification command. The appendix and manuscript are separate deliverables; neither depends on access to that history for its mathematical content.